Final Project Report – Team 78

(Project Title : IntellInvest in Airbnb)

(Team Member : Heba Maghraby, James Lai, Qi (Sophie) Pan, Sanjit Paikaray, Timothy Haixiang Zhang)

Contents

1 Project Objective **2**
- 1.1 Background of Project . . . 2
- 1.2 Project Topic : Predict the location/city for highest investment return in Airbnb industry for the investors . . . 2

2 Overview of the Project **2**
- 2.1 Problem Statement . . . 2
- 2.2 Initial Hypotheses . . . 2

3 Data Sets **2**
- 3.1 Data Sources . . . 2
- 3.2 Data Description . . . 2
  - 3.2.1 Airbnb data : . . . 2
  - 3.2.2 Zillow data : . . . 3

4 Approach and Methodology **4**
- 4.1 Data Pre-Processing and Feature Selection . . . 4
  - 4.1.1 Airbnb Data : . . . 4
  - 4.1.2 Zillow Data : . . . 4
  - 4.1.3 Feature Selection : . . . 4
  - 4.1.4 ROI(Return on Investment) : . . . 4
- 4.2 EDA (Exploratory Data Analysis) . . . 4
  - 4.2.1 Airbnb Data Set : . . . 4
  - 4.2.2 Zillow Data Set : . . . 6
- 4.3 ROI Calculation . . . 6
- 4.4 Data Model . . . 7
  - 4.4.1 Linear Regression . . . 7
  - 4.4.2 Gradient Boosted Tree Model (GBT) . . . 10

5 Model Results Comparison **11**
- 5.1 Which model family generalizes better? . . . 11
- 5.2 Do the models highlight the same important features? . . . 12

6 Area of future scope **12**

7 Conclusion **12**

8 Meet the Team **12**

9 References **12**

10 Appendix **13**

1 Project Objective

1.1 Background of Project

In 2008 Brian Chesky, Joe Gebbia, and Nathan Blecharczyk founded the company Airbnb to help tourists find short term rentals and support homeowners who would like to make an extra income by renting out their homes or even a room within their homes. With the tourism industry flourishing again after 2 years of quarantine travel restriction due to COVID, investors are investigating the idea of purchasing a house to turn it into an Airbnb rental.

1.2 Project Topic : Predict the location/city for highest investment return in Airbnb industry for the investors

We aim to help investors decide how to spend their money wisely and identify where to buy real estate for short term rentals as well as determine which location has the highest investment return. Our goal is to identify the key factors that contribute to a listing price so that Airbnb hosts understand how they can set up their investment for maximum impact as well as determine if the returns are worthwhile. Cities expected to perform well in this investment are selected based on the emerging trends in real states report by pwc(Refer to the Reference section in the document for the details).

2 Overview of the Project

2.1 Problem Statement

- The goal of this project is to identify the key factors that contribute to a listing price so that Airbnb hosts understand how they can set up their investment for maximum impact as well as determine if the returns are worthwhile
- Predict the ranking of the cities from highest return investment in Airbnb Industry

2.2 Initial Hypotheses

- The expectation is that a global model will be better at predicting prices for cities which are not represented in the data set
- For any city which has a similar counterpart in the data set, it should be expected that the local counterpart will be a better representative model
- With respect to the feature analysis, it should be expected that the same features are important across markets using the same model type

3 Data Sets

3.1 Data Sources

1. Airbnb Listing and Calendar Dataset: `http://insideairbnb.com/get-the-data/`
2. Zillow Dataset: `https://www.zillow.com/research/data/`

3.2 Data Description

Altogether we are using total 3 data sets in our project. 2 data sets (Airbnb Listing and Airbnb Calendar) from Airbnb website and one more additional data set from Zillow data source.

3.2.1 Airbnb data :

For this dataset, we retrieved October 2021 data for all 13 cities (Figure 1: List of 13 cities used in mode) to have a good comparison for tuning the models as well as comparing the results.

Training Cities	Testing Cities
Dallas	Austin
Chicago	San Diego
Los Angeles	Newark
Columbus	
Denver	
San Francisco	
New York City	
Seattle	
Portland	
Boston	

Figure 1: List of 13 cities used in model

- **Listing Data set:**
  - Listing data set contains information about each listing with listing id, price, number of rooms, type of house, review rating etc. The data is broken up by cities, so we can consider the each individual city data for each local trained model and one combine multiple cities data for global model.It contained around 74 variables.
  - The listing data set for each city contains good amount of records. For reference, Dallas city for October, 2021 contains around 4461 records. For details refer to (Figure 21: Data set statistics) in Appendix section.
- **Calendar Data set:**
  - Calendar data set contains listing id, price and dates for which the property was reserved as well as availability for future dates.It contained around 7 variables.
  - For reference, Dallas city for October, 2021 contains around 1627888 records. Also we can see same number of records for other cities in the Airbnb data set.or details refer to (Figure 21: Data set statistics) in Appendix section.
- **Key Variables:** As per the initial analysis, we identified some of the important features which was listed below from both the data sets(Figure 2: Key Variables).

Field	Type	Description
id	integer	Airbnb's unique identifier for the listing
neighbourhood	text	Neighbouhood details
latitude	numeric	Uses the World Geodetic System (WGS84) projection for latitude and longitude.
longitude	numeric	Uses the World Geodetic System (WGS84) projection for latitude and longitude.
property_type	text	Self selected property type. Hotels and Bed and Breakfasts are described as such by their hosts in this field
room_type	text	[Entire home/apt|Private room|Shared room|Hotel]
accommodates	integer	The maximum capacity of the listing
bathrooms	numeric	The number of bathrooms in the listing
price	currency	daily price in local currency
minimum_nights	integer	minimum number of night stay for the listing (calendar rules may be different)
maximum_nights	integer	maximum number of night stay for the listing (calendar rules may be different)
review_scores_rating	numeric	Review score Rating
review_scores_accuracy	numeric	Review score accuracy

Figure 2: Key Variables

3.2.2 Zillow data :

- Dataset contains Zillow Home Value Index (ZHVI): A smoothed, seasonally adjusted measure of the typical home value and market changes across a given region and housing type. It reflects the typical value for homes in the 35th to 65th percentile range.
- Data selected contain data for single family residence & Condos.
- The data set has 22,440 records and columns describing region name, region type, and state name. Refer to (Figure 22: Zillow Data set) in Appendix section for sample records of the data set.

4 Approach and Methodology

4.1 Data Pre-Processing and Feature Selection

4.1.1 Airbnb Data :

For our analysis and modelling, we collected the Airbnb data only for October 2021 for all 13 cities from both listing as well as calendar data set. Categorical variables such as super host, neighborhood and numeric variables such as bedrooms, listing count contained missing values which could not be imputed and could give incorrect results. So respective rows were removed containing missing values.Couple of variables like price, bathroom_text contained '$' and other unwanted characters which were removed and converted into Float variable.Depending on the ID field, both the data sets such as listing as well as calendar data sets were merged. Additional field "Income" was derived by aggregating of the price from Calendar data set for all ids booked in the past. Rows related to Hotel_room was removed from the data set as it was not applicable for the modelling. (Figure 23: Merged cleaned Airbnb Data Set) in Appendix section).

4.1.2 Zillow Data :

Zillow data was filtered for Texas, Dallas. And merged with the Airbnb data ROI % was calculated (Figure 24: Merged zillow and Airbnb data set) in Appendix section.

4.1.3 Feature Selection :

From the merged data set we removed the unwanted columns or independent variables as per our research till now. As later stage, we are going use some more feature selection techniques like forward and backward elimination, and lasso technique.

4.1.4 ROI(Return on Investment) :

As our project objective is to come up with ranking of selected cities available in Airbnb site from investment perspective in Airbnb. Therefore, we will derive Annual ROI for all cities after our model price prediction. This would be a metric that we would calculate based on the historical bookings of individual properties and the average housing prices for a given market. ROI will be calculated only for full house types.
Equation used for ROI is as follow:

$$ROI\% \text{ monthly} = \left(\frac{\frac{houseprice - (houseprice * 20\% downpayment) + (5\% interestrate)}{20 years}}{12} \right) - (1\% \text{ property } \frac{tax}{12} \text{ months })$$

4.2 EDA (Exploratory Data Analysis)

4.2.1 Airbnb Data Set :

1. **Airbnb price changes on monthly basis**
   We could see that October month where the price was high.We assume that during October month the COVID-19 risk was little low due to which there might be more demand in October month.(Figure 3: Airbnb price changes Room type popularity)

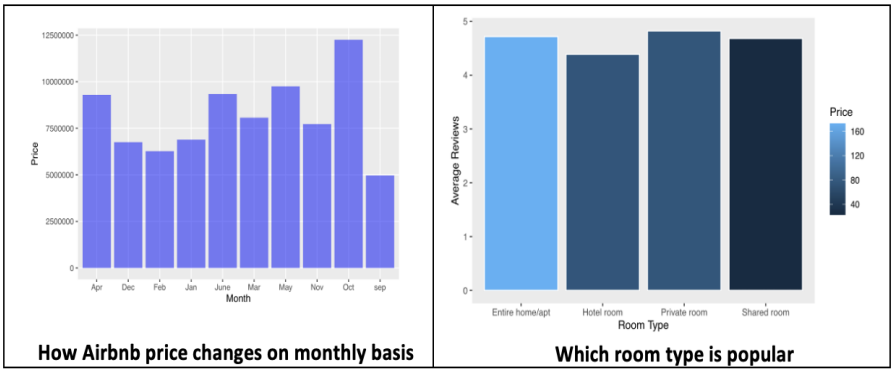


Figure 3: Airbnb price changes Room type popularity

2. **Room Type Popularity**
   Entire home type is most expensive where shared room is cheaper which is making sense. Both entire home as well as private room has better review rating (Figure 3: Airbnb price changes Room type popularity)

3. **Room Type Analysis**
   We would expect the listings that are private rooms or shared rooms to be from homeowners renting out spare rooms while entire home/apt is more likely to be real estate investors trying to generate more profit. It seems like some cities, such as Boston, are more accommodating to homeowners. However, the majority of Airbnb hosts are by far real estate investors. This suggests that this is a prime opportunity to investigate real estate purchases for short-term rentals (Figure 4: Room Type Analysis Correlation Plot Analysis)

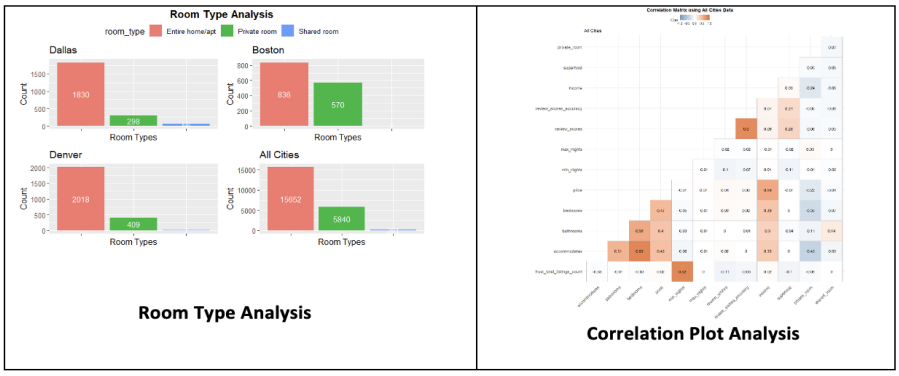


Figure 4: Room Type Analysis Correlation Plot Analysis

4. **Correlation Plot Analysis**
   Based on the correlation matrix, we saw a few variables that are highly correlated with each other. Most of the high correlations seem reasonable. One interesting thing to note is the very high correlation between the number of bathrooms with the price. This seems in line with a ratio of the number of bedrooms to the number of bathrooms that a property usually has, so it could be a variable that may cause some multicollinearity. The variables with negative correlations are also expected. The main two variables with negative correlations are private_room and shared_room, which are expected. These room types are generally associated with a single room out of an entire property, so it makes sense that it would accommodate less people and be priced less(Figure 4: Room Type Analysis Correlation Plot Analysis).

5. **Accommodates versus Price**
   Overall, it looks like there is increasing trend where the number of people that a listing accommodates leads to a higher listing price. Again, we see some significant outliers in terms of listing pricing where some listings are more than triple the price of similarly sized listing. This would be an interesting to explore that differentiates these types of properties whether they are in a more desirable locations or with better amenities (Figure 5: Accommodates versus price Super Host Analysis).

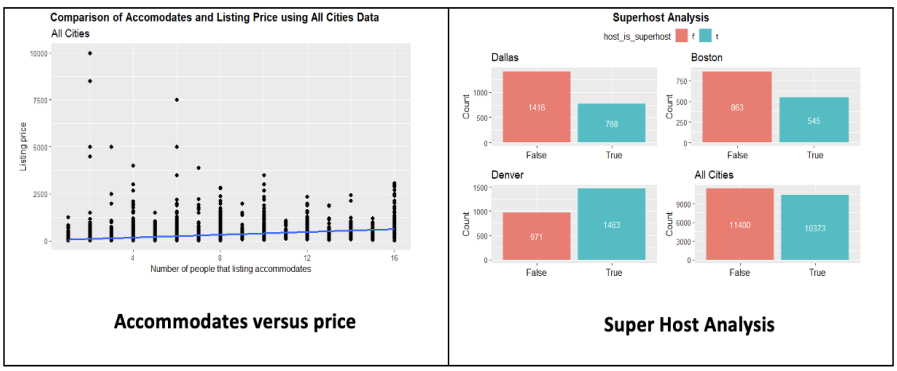


Figure 5: Accommodates versus price Super Host Analysis

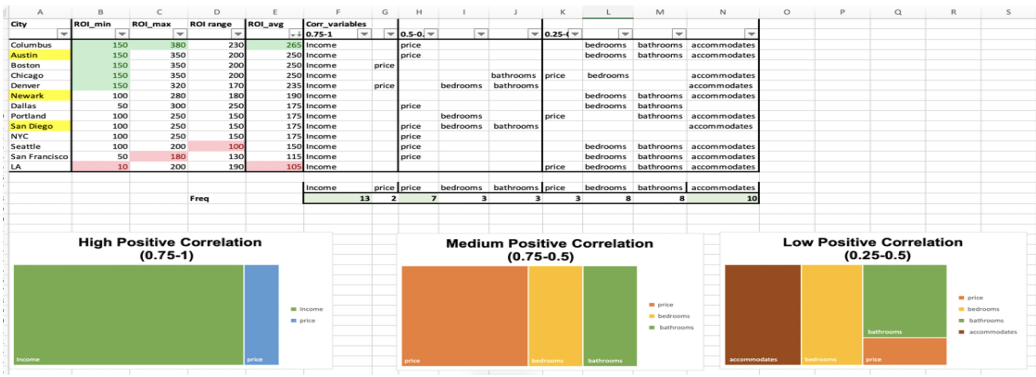
6. **Superhost Analysis**
   Superhost is a designation from Airbnb given to hosts that met their criteria: having high overall ratings,

over 10+ stays, low cancellation rates, and high response rates. Airbnb advertises Superhosts as having greater visibility and earning potential, but what we see from the data is there is significant number of superhosts throughout the platform. This varies between cities, but overall a large proportion of hosts are actually superhosts. With so many users as superhosts, the financial value of being a superhost may be diluted and may reduce any price premiums that they may previously been able to charge (Figure 5: Accommodates versus price Super Host Analysis).

4.2.2 Zillow Data Set :

1. **ROI on Zillow Data :**
   EDA for ROI gave the below results:

   - ROI monthly percentages have a wide range from 0% to 5000%.
   - The majority of ROI % fall between 200% and 400%
   - ROI% showed a strong correlation with house price, which is expected as ROI% is derived from house price
   - ROI% also showed moderate correlation with bedrooms and accommodates.
   - Graphs are available in (Figure 6: ROI EDA - Part 1 and Figure 7: ROI EDA - Part 2)



City	ROI_min	ROI_max	ROI range	ROI_avg	Corr_variables 0.75-1		0.5-0.			0.25-(			
Columbus	150	380	230	265	Income		price				bedrooms	bathrooms	accommodates
Austin	150	350	200	250	Income		price				bedrooms	bathrooms	accommodates
Boston	150	350	200	250	Income	price							
Chicago	150	350	200	250	Income				bathrooms	price	bedrooms		accommodates
Denver	150	320	170	235	Income	price		bedrooms	bathrooms				accommodates
Newark	100	280	180	190	Income						bedrooms	bathrooms	accommodates
Dallas	50	300	250	175	Income		price				bedrooms	bathrooms	
Portland	100	250	150	175	Income			bedrooms		price		bathrooms	accommodates
San Diego	100	250	150	175	Income		price	bedrooms	bathrooms				accommodates
NYC	100	250	150	175	Income		price						
Seattle	100	200	100	150	Income		price				bedrooms	bathrooms	accommodates
San Francisco	50	180	130	115	Income		price				bedrooms	bathrooms	accommodates
LA	10	200	190	105	Income					price	bedrooms	bathrooms	accommodates
					Income	price	price	bedrooms	bathrooms	price	bedrooms	bathrooms	accommodates
			Freq		13	2	7	3	3	3	8	8	10

Figure 6: ROI EDA - Part 1

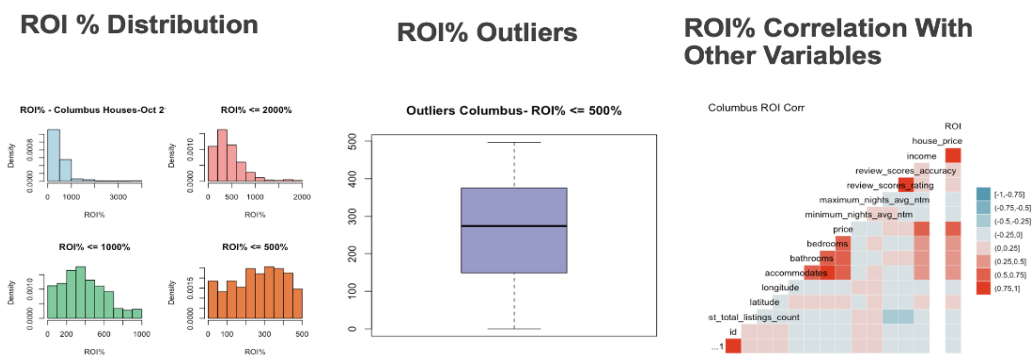


Figure 7: ROI EDA - Part 2

4.3 ROI Calculation

We used a subset of each city's data to calculate the city's ROI%. The subset included rentals featuring entire houses or apartments to be compatible with the Zillow data. Then, we performed EDA on all cities ROI% separately for the month of October 2021. The below findings were observed:

- All ROI histograms were skewed to the right
- Boxplots indicated presence of outliers
- Most relevant ROI distributions were for ROI% <= 500%

- Features that showed most correlations with ROI% are Price, Bedrooms Bathrooms
- The city with highest average ROI is Columbus with 265%
- The city with lowest average ROI is Los Angeles with 105%
- The cities ranked from highest average ROI% to Lowest average ROI% are as below:

Cities Ranked From Highest Average ROI% to Lowest Average ROI%	Average ROI %
Columbus	265%
Boston	250%
Chicago	250%
Denver	235%
Dallas	175%
Portland	175%
New York City	175%
Seattle	150%
San Francisco	115%
Los Angeles	105%

Figure 8: ROI EDA

4.4 Data Model

4.4.1 Linear Regression

We selected 11 variables from the total 73 variables to build linear regression models with 'price' as the dependent variable. After checking the distribution of each variable in each city and the whole US dataset, none of them is normally distributed which indicates log transformation might be a good try later. For collinearity between variables, there are several pairs of predictors with correlation bigger than 0.7 (eg. review_scores and review_scores_accuracy, bedrooms and accommodates). So variable selection is necessary. For categorical variables, we created dummy variables and use 'not-superhost' and the 'Entire home/apt' to the base cases. After separating the dataset into 30% testing and 70% training, we built linear-linear, linear-log, log-linear, and log-log models and three more lasso regression models.

This methodology has been applied to all the city level and the US level models.

After comparing all 9 models' performance by Adjusted R-squared, AIC, RMSE (Root Mean Squared Error) on cross-validation, and RMSE on the test dataset. The log-linear model with all 11 variables is final model for all the city level and the US level models.

Here is the performance chart of all Dallas models(Figure 9: Performance chart of Dallas models):

Model	Number predictors	R squared	Adjusted R squared	AIC	RMSE by DAAG CV	RMSE on test dataset
model1_linear_linear	11	0.5394	0.536	18686.57	110	105.65
model2_linear_linear	7	0.5372	0.5351	18685.57	110	105.96
model3_log_log	11	0.6796	0.6773	1570.61	0.406	0.404
model4_log_log	8	0.6787	0.677	1568.98	0.405	0.406
model5_log_linear	11	0.7023	0.7	1493.65	0.396	0.398
model6_linear_log	11	0.432	0.428	18888.37	117	118
lasso_cv_linear_linear	2	0.433	na	na	122	113
lasso_cv_log_log	9	0.663	na	na	0.415	0.414
lasso_cv_log_linear	10	0.687	na	na	0.405	0.408

Figure 9: Performance chart of Dallas models

- Choose from the top 2 models (model5_log_linear, model4_log_log) by regression diagnostics
- Final model: model5_log_linear

Details of the final Dallas model(Figure 10: Final Dallas model) :

```
##
## Call:
## lm(formula = price_log ~ ., data = train_dallas1_logprice)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.07821 -0.25327 -0.03015  0.19379  2.11666
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)               3.738e+00  1.479e-01  25.269  < 2e-16 ***
## host_total_listings_count 5.409e-04  1.381e-04   3.916 9.42e-05 ***
## accommodates              4.487e-02  6.347e-03   7.070 2.36e-12 ***
## bathrooms                 2.167e-01  2.026e-02  10.695  < 2e-16 ***
## bedrooms                  1.215e-01  2.295e-02   5.295 1.37e-07 ***
## min_nights               -3.702e-03  8.157e-04  -4.538 6.14e-06 ***
## max_nights               -7.412e-05  2.602e-05  -2.849  0.00445 **
## review_scores             1.769e-01  4.502e-02   3.929 8.93e-05 ***
## review_scores_accuracy   -8.028e-02  4.652e-02  -1.726  0.08458 .
## superhost                 6.869e-02  2.286e-02   3.004  0.00271 **
## private_room             -6.192e-01  3.215e-02 -19.258  < 2e-16 ***
## shared_room              -1.634e+00  6.915e-02 -23.636  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3927 on 1516 degrees of freedom
## Multiple R-squared:  0.7023, Adjusted R-squared:  0.7001
## F-statistic: 325.1 on 11 and 1516 DF,  p-value: < 2.2e-16
```

Figure 10: Final Dallas model

The top 5 important features of the final Dallas model(Figure 11: Top 5 important features of the final Dallas mode) :

- shared_room(-1.634e+00)
- private_room(-6.192e-01)
- bathrooms(+2.167e-01)
- review_scores(+1.769e-01)
- bedrooms(+1.215e-01)

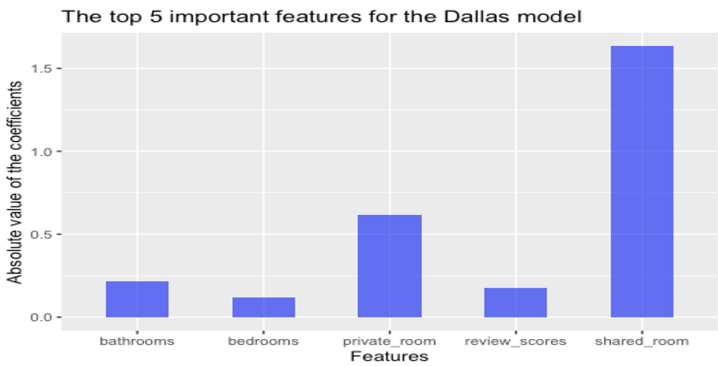


Figure 11: Top 5 important features of the final Dallas model

Here is the performance chart of the 10 US cities' models:

Model	R squared	Adjusted R squared	AIC
Boston_log_linear_model	0.623	0.62	1990
chicago_log_linear_model	0.594	0.593	5553
newyork_log_linear_model	0.494	0.494	24017
la_log_linear_model	0.629	0.629	21751
dallas_log_linear_model	0.698	0.696	2145
columbus_log_linear_model	0.671	0.667	733
portland_log_linear_model	0.517	0.515	2167
sfo_log_linear_model	0.547	0.546	4647
denver_log_linear_model	0.529	0.527	3237
seattle_log_linear_model	0.615	0.613	2377

Figure 12: Performance chart of the 10 US cities' models

- The best model is Dallas
- The worst model is New York City

Note: the Adjusted R-squared for Dallas model here is a little different from the model5_log_linear is because after choosing the best model type, we retrain the model with the whole Dallas dataset.

From above we can see the avg adjusted R-squared is 0.59. An interesting found is with the data set getting noisier, the R-squared or the performance of the model is worse. For example, the dataset for New York City is the biggest. We all know New York is a real 'Jungle'. The range of the rental price, and the gap of income difference within different areas are huge. These factors all contribute to its model's deficient performance(Figure 12: Performance chart of the 10 US cities' models).

Here is the performance chart of the US-level models(Figure 13: Performance chart of the US-level models):

Model	Number predictors	R squared	Adjusted R squared	AIC	RMSE by DAAG CV	RMSE on test dataset
model1_US_linear_linear	11	0.3142	0.3137	201603.5	181	226.165
model2_US_linear_linear	7	0.3134	0.3131	201613.1	181	226.26
model3_US_log_log	11	0.5129	0.5126	22093.42	0.501	0.505
model4_US_log_log	7	0.5128	0.5126	22089.03	0.5	0.505
model5_US_log_linear	11	0.522	0.5217	22135.84	0.5	0.507
model6_US_linear_log	11	0.2401	0.2396	201961.5	186	227.06
lasso_cv_US_linear_linear	3	0.234	na	na	191	228
lasso_cv_US_log_log	8	0.507	na	na	0.504	0.508
lasso_cv_US_log_linear	10	0.516	na	na	0.503	0.509

Figure 13: Performance chart of the US-level models

- Choose from the top 2 models (model4_US_log_log, model5_US_log_linear) by regression diagnostics
- Final model: model5_US_log_linear

Details of the final US model(Figure 14: Details of the final US model):

```
##
## Call:
## lm(formula = price_log ~ ., data = train_US1_logprice)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.8094 -0.3348 -0.0488  0.2751  4.1706
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)             3.873e+00  6.392e-02  60.594  < 2e-16 ***
## host_total_listings_count 5.095e-04  5.755e-05   8.854  < 2e-16 ***
## accommodates            4.644e-02  2.953e-03  15.727  < 2e-16 ***
## bathrooms               1.456e-01  6.608e-03  22.035  < 2e-16 ***
## bedrooms                1.881e-01  7.995e-03  23.522  < 2e-16 ***
## min_nights             -6.776e-04  1.423e-04  -4.761 1.95e-06 ***
## max_nights              5.344e-10  8.823e-11   6.057 1.42e-09 ***
## review_scores           1.910e-01  1.967e-02   9.709  < 2e-16 ***
## review_scores_accuracy -9.616e-02  2.085e-02  -4.612 4.03e-06 ***
## superhost               4.588e-03  8.520e-03   0.539     0.59
## private_room           -5.143e-01  1.043e-02 -49.322  < 2e-16 ***
## shared_room            -1.246e+00  3.662e-02 -34.041  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5 on 15229 degrees of freedom
## Multiple R-squared:  0.522,  Adjusted R-squared:  0.5217
## F-statistic:  1512 on 11 and 15229 DF,  p-value: < 2.2e-16
```

Figure 14: Details of the final US model

The top 5 important features for the final US model(Figure 15: Top 5 important features for the final US model):

- shared_room(-1.246e+00)
- private_room(-5.143e-01)
- review_scores(+1.910e-01)
- bedrooms(+1.881e-01)
- bathrooms(+1.456e-01)

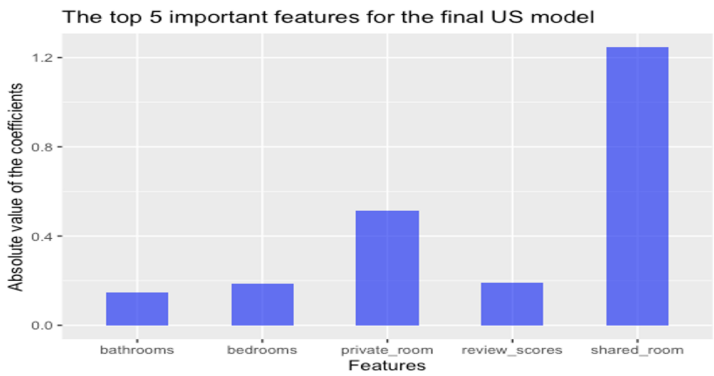


Figure 15: Top 5 important features for the final US model

The rank of the top 5 important features for the US model is a little different from the Dallas model. But the features themselves are the same.

Performance Comparison for US Level Model vs City Level Model(Figure 16: Comparison for US Level Model vs City Level Model):

- Use the best Dallas model to test Austin vs using the best US model to test Austin
- Use the best Los Angeles model to test San Diego vs using the best US model to test San Diego
- Use the best New York City model to test Newark vs using the best US model to test Newark

Test City	City Level Model	City Level Model RMSE	US Level Model RMSE	Winner Model
Austin	dallas_log_linear_model	0.62	0.558	US model
San Diego	la_log_linear_model	0.45	0.478	city level model
Newark	newyork_log_linear_model	0.718	0.679	US model

Figure 16: Comparison for US Level Model vs City Level Model

Conclusion for the linear regression method:

- The Log-linear model is the best choice for the linear regression method for both the city level and the whole US level
- Normally, the noisier the data set, the worse the model's performance. (New York City model vs Dallas model: adjusted R-squared: 0.494 vs 0.696)
- When predicting Airbnb rental prices for some random city, it is uncertain which model is better: the US model or the neighbor city model. Since it also depends on the similarity between the neighbor city and the target city. For example, although Newark and New York City are next to each other, their employment environment, salary levels, and real estate markets are quite different. In this situation, the US model performs better than the city model. In contrast, if both cities are quite similar like Los Angeles and San Diego, the city model will perform better than the US model.

4.4.2 Gradient Boosted Tree Model (GBT)

We used the popular CatBoost Python library for our GBT implementation and followed the same experimental setup as linear regression. Specifically, we trained both local models (Dallas, LA, New York) and a global model and evaluated the generalization capacity of the models against a set of similar test cities (Austin, San Diego, Newark).

Hyperparameter Search

We restricted the hyperparameter search space to 3 of the most important levers: learning rate, iterations, and depth. The depth controls the maximum depth of an individual tree and is important to control overfitting. The iterations controls the number of trees which are additively appended to the ensemble. Finally, the learning

rate is analogous to other models and controls the rate at which the model changes parameters based on the current loss.

Model Performance

GBT had mixed results with respect to different cities as shown in (Figure 17: GBT Training Metrics). In some cities, such as Dallas and LA, GBT outperformed linear regression in terms of R^2, while in others it did poorly.

	model	rmse	r2
0	Dallas	75.081864	0.767992
2	Los Angeles	205.921269	0.716196
4	New York	204.245133	0.270105
6	Global	195.893835	0.528817

Figure 17: GBT Training Metrics

(Figure 18: GBT Test Metrics) shows that GBT was not able to generalize well to other cities. We posit that this is due to the non-linear model overfitting to the training cities.

	model	rmse	r2
1	Dallas vs Austin	453.852147	0.098030
3	Los Angeles vs San Diego	200.445448	0.470456
5	New York vs Newark	314.922904	-0.159525
7	Global vs Austin	426.197506	0.204601
8	Global vs San Diego	196.443860	0.491388
9	Global vs Newark	314.922904	-0.159525

Figure 18: GBT Test Metrics

Feature Importance

(Figure 19: GBT Feature Importances) shows the feature importances for the global GBT model. Here we see that bathrooms, bedrooms, and room type are some of the most important features. For the full set of feature importances, refer to our GBT implementation.

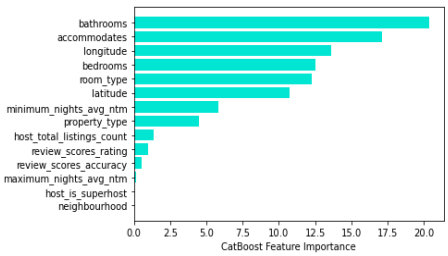


Figure 19: GBT Feature Importances

5 Model Results Comparison

5.1 Which model family generalizes better?

- From our comparison of GBT vs Linear Regression, it seems that the simpler linear regression model generalizes better across markets.
- GBT was able to achieve strong R2 but was not able to generalize well to new markets due to overfitting.

5.2 Do the models highlight the same important features?

- Both models highlighted a similar set of features as most important (bedrooms, bathrooms, room type).
- Linear Regression focused on Room Type as the most important feature.
- GBT focused more on bedrooms and bathrooms.

6 Area of future scope

There are several improvements can be done for the future while predicting the best price and also coming up with ranking for ROI.

- More Data collection - For our project, we collected the data only for 13 cities and it was for October 2021. We can go for more city as well as more data in future.
- Feature Engineering and Selection - We can add some more additional features like calculation of nearby cities by considering the longitude and latitude and additional feature engineering (e.g. KNN features)
- Machine Learning Model - For this project, we considered Linear Regression model and GBT. But in future, we can consider Deep learning for the prediction

7 Conclusion

In conclusion, we wanted to recap our findings against our initial hypothesis. For our first two hypotheses, we were looking at the scenarios where the global or the individual city models would perform better. Based on our results, the answer varied depending on which type of model we were using. With the regression model, neither model consistently outpredict each other, so investors would need to evaluate cities on a case by case basis. However, with the GBT model, the global model did perform better than the local counterparts. For our third hypothesis, we were interested in feature analysis and expected the same features to be important across the different models. This held true as the strongest features between the models were the bedrooms, bathrooms, and room type.

8 Meet the Team

Sophie Pan | Sanjit Paikaray | Tim Zhang | Heba Maghraby | James Lai

Figure 20: Meet the Team

9 References

1. Emerging Trends in Real Estate 2022 : `https://knowledge.uli.org/-/media/files/emerging-trends/2022/emerging-trends-in-real-estate-united-states-and-canada---2022.pdf?rev=2d5130961f914eff97538`

10 Appendix

- **Team-78 GitHub details :** `https://github.gatech.edu/MGT-6203-Summer-2022-Canvas/Team-78.git`
- **Data set statistics :**

Training City			
City	Listing Data Set Record Count	Calendar Data Set Record Count	Filtered Listing and Calendar Merged Data set Record Count
Boston	3123	1139165	1408
Columbus	1351	493115	925
Dallas	4461	1627888	2184
Denver	3850	1404859	2434
Portland	3540	1290655	2357
Seattle	4249	1549846	2465
Chicago	6455	2355772	3819
SFO	6616	2414113	3434
Los Angels	32421	11832398	14465
New York	36923	13470765	16202
Test City			
Newark	894	326310	430
Austin	10809	3944923	5214
San Diego	9672	3529957	5682

Figure 21: Data set statistics

- **Zillow Dataset :**

RegionID	SizeRank	RegionName	RegionType	StateName	State	Metro	CountyName	2000-01-31
6181	0	New York	City	NY	NY	New York-Newark-Jersey City	Queens County	253326.0
12447	1	Los Angeles	City	CA	CA	Los Angeles-Long Beach-Anaheim	Los Angeles County	238138.0
39051	2	Houston	City	TX	TX	Houston-The Woodlands-Sugar Land	Harris County	109646.0
17426	3	Chicago	City	IL	IL	Chicago-Naperville-Elgin	Cook County	169437.0
6915	4	San Antonio	City	TX	TX	San Antonio-New Braunfels	Bexar County	116294.0
13271	5	Philadelphia	City	PA	PA	Philadelphia-Camden-Wilmington	Philadelphia County	74647.0
40326	6	Phoenix	City	AZ	AZ	Phoenix-Mesa-Scottsdale	Maricopa County	124357.0
18959	7	Las Vegas	City	NV	NV	Las Vegas-Henderson-Paradise	Clark County	165493.0
54296	8	San Diego	City	CA	CA	San Diego-Carlsbad	San Diego County	231501.0
38128	9	Dallas	City	TX	TX	Dallas-Fort Worth-Arlington	Dallas County	121454.0

Figure 22: Zillow Data set

- **Merged cleaned Airbnb Data Set :**

	id	host_is_superhost	host_total_listings_count	neighbourhood	latitude	longitude	property_type	room_type	accommodates	bathrooms_text	bedrooms	price	minimum_nights_avg_ntm	maximum_nights_avg_ntm	review_scores_rating	review_scores_accuracy	income
0	61878	t	1	Dallas, Texas, United States	32.81829	-96.81893	Entire condominium	Entire home/apt	3	1 bath	1	$72.00	30	1125	4.75	4.82	1707
1	270936	f	2	Dallas, Texas, United States	32.81442	-96.81548	Entire condominium	Entire home/apt	2	1 bath	1	$75.00	30	90	4.93	4.95	2325
2	776810	t	1	Dallas, Texas, United States	32.81304	-96.8154	Entire loft	Entire home/apt	2	1.5 baths	1	$75.00	5	1125	4.9	4.89	1350
3	795703	t	2	Dallas, Texas, United States	32.80158	-96.81117	Entire condominium	Entire home/apt	3	1 bath	1	$199.00	30	1125	4.84	4.91	1206
4	860248	f	4	Dallas, Texas, United States	32.85521	-96.70482	Private room in	Private room	2	1 private bath	1	$38.00	3	365	4.84	4.82	92
5	1056850	f	3	Dallas, Texas, United States	32.82057	-96.80932	Entire rental unit	Entire home/apt	2	1 bath	1	$95.00	2	1125	4.98	5	855
6	1158488	t	2	Dallas, Texas, United States	32.81696	-96.76733	Entire guest suite	Entire home/apt	2	1 bath	1	$99.00	3	1125	4.97	4.97	695
7	1277933	f	2	Dallas, Texas, United States	32.92532	-96.79028	Private room in	Private room	4	1 shared bath	2	$65.00	1	1125	4.63	4.7	2015
8	1479800	t	2	Dallas, Texas, United States	32.83141	-96.76348	Private room in	Private room	2	1 private bath	1	$80.00	2	1125	4.93	4.95	640
9	1488809	t	1	Dallas, Texas, United States	32.74605	-96.83386	Private room in	Private room	2	1 bath	1	$55.00	1	28	4.91	4.96	1705
10	1774599	f	5	Dallas, Texas, United States	32.80869	-96.73121	Entire rental unit	Entire home/apt	2	1 bath	1	$40.00	24	1125	4.42	4.62	840
11	1826550	t	3	Dallas, Texas, United States	32.78112	-96.79165	Private room in	Private room	3	1 private bath	1	$82.00	1	1125	4.94	4.97	1275

Figure 23: Merged cleaned Airbnb Data Set

– **Merged Zillow and Airbnb Data Set :**

	id	host_is_superhost	host_total_listings_count	neighbourhood	latitude	longitude	property_type	room_type	accommodates	bathrooms_text	bedrooms	price	minimum_nights_avg_ntm	maximum_nights_avg_ntm	review_scores_rating	review_scores_accuracy	income	house_price	ROI
1	61878	TRUE	1	Dallas, Texas, United States	32.81829	-96.81869	Entire condominium (condo)	Entire home/apt	3	1 bath	1	$72.00	30	1125	4.75	4.62	1707	285393	221
2	270936	FALSE	2	Dallas, Texas, United States	32.81442	-96.81548	Entire condominium (condo)	Entire home/apt	2	1 bath	1	$75.00	30	90	4.93	4.95	2325	285393	301
3	776810	TRUE	1	Dallas, Texas, United States	32.81304	-96.8154	Entire loft	Entire home/apt	2	1.5 baths	1	$75.00	5	1125	4.9	4.89	1350	285393	175
4	796703	TRUE	2	Dallas, Texas, United States	32.80158	-96.81117	Entire condominium (condo)	Entire home/apt	3	1 bath	1	$199.00	30	1125	4.84	4.81	1206	285393	156
5	1356850	FALSE	3	Dallas, Texas, United States	32.82057	-96.80932	Entire rental unit	Entire home/apt	2	1 bath	1	$95.00	2	1125	4.98	5	855	285393	111
6	1774399	FALSE	5	Dallas, Texas, United States	32.80869	-96.73121	Entire rental unit	Entire home/apt	2	1 bath	1	$40.00	24	1125	4.42	4.62	840	285393	109
7	1893437	TRUE	3	Dallas, Texas, United States	32.77865	-96.79962	Entire townhouse	Entire home/apt	8	3.5 baths	3	$370.00	3	1125	4.99	4.68	5396	285393	698

Figure 24: Merged Zillow and Airbnb Data Set